

Component Material Balance Equation

Darcy's law governs the flow in the reservoir. The dispersion and diffusion of components in the aqueous phase also contribute to the movement of components in the aqueous phase. Let n_{ct} be the total number of species. The components are ordered as follows: n_c gaseous components, n_a aqueous components and n_m mineral components. The conservation equations for the different species are:

$$\begin{aligned} \psi_i \equiv & \Delta T_g^u y_{ig}^u (\Delta p^{n+1} + \Delta P_{cw}^u - \tilde{\rho}_g^u g \Delta d) \\ & + \Delta T_w^u y_{iw}^u (\Delta p^{n+1} - \tilde{\rho}_w^u g \Delta d) + \sum_{q=g,w} \Delta D_{iq}^u \Delta y_{iq}^u \end{aligned} \quad (D10.1)$$

$$+ V \sigma_{i,aq}^{n+1} + q_i^{n+1} - \frac{V}{\Delta t} (N_i^{n+1} - N_i^n) = 0, \quad i = 1, \dots, n_c$$

$$\begin{aligned} \psi_j \equiv & \Delta T_w^u y_{jw}^u (\Delta p^{n+1} - \tilde{\rho}_w^u g \Delta d) + \Delta D_{iw}^u \Delta y_{iw}^u + V \sigma_{j,aq}^{n+1} + V \sigma_{j,mn}^{n+1} \\ & + q_j^{n+1} - \frac{V}{\Delta t} (N_{ja}^{n+1} - N_{ja}^n) = 0 \quad j = n_c + 1, \dots, n_c + n_a \end{aligned} \quad (D10.2)$$

$$\psi_k \equiv V \sigma_{k,mn}^{n+1} - \frac{V}{\Delta t} (N_k^{n+1} - N_k^n) = 0, \quad k = n_c + n_a + 1, \dots, n_{ct} \quad (D10.3)$$

In the above equation, the superscript n denotes the old time level and $n+1$ denotes the new time level. Equations (D10.1) and (D10.2) are discretized in an adaptive-implicit manner (Nghiem and Rozon, 1988 and 1989, and Collins et al, 1992). The superscript $u = n$ for explicit grid blocks and $n+1$ for implicit grid blocks. The terms $(V \sigma_{i,aq})$ and $(V \sigma_{i,mn})$ in Equations (D10.1) and (D10.2) correspond respectively to the intra-aqueous reaction rates and mineral dissolution/precipitation rates. $\sigma_{i,mn}$ is related to the rates in Equation (D8.1) as follows:

$$V \sigma_{i,mn} = V S_w \sum_{\beta=1}^{R_{mn}} r_{i\beta} = V S_w \sum_{\beta=1}^{R_{mn}} \nu_{i\beta} r_{\beta} \quad (\text{D10.4})$$

As r_{β} [mol/(m³ s)] is the reaction rate per unit bulk volume of rock when the pore space is filled with water, multiplication by the factor VS_w is required in the above equation to obtain the proper rate when part of the pore space is occupied by gas. The $(V\sigma_{i,aq})$ correspond to rates of the chemical equilibrium reactions. There is no accurate means available to compute these rates, but they can be eliminated as described in the following section.